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1 Power Query Editor

Power Query Editor is the data preparation and transformation engine in Microsoft Power BI. It enables users to connect to multiple data sources, clean and reshape data, combine datasets, remove inconsistencies and prepare high-quality data before loading it into the Power BI data model.

Almost every professional Power BI project begins with Power Query because clean and structured data forms the foundation of reliable reports, dashboards, and DAX calculations.

This chapter introduces the Power Query Editor interface, its components, common transformations, query management, and best practices used in enterprise Business Intelligence (BI) projects.

Definition:

Power Query Editor is Microsoft's Extract, Transform, and Load (ETL) tool available in Power BI Desktop. It enables users to prepare raw data before it is loaded into the data model.

Power Query follows the ETL process:

- Extract data from one or more sources.
- Transform the data.
- Load the cleaned data into Power BI.

Power Query records every transformation as a reusable step, allowing automatic execution during every refresh.

Problems

1.1 Open the Power Query Editor for an imported Sales dataset and identify its major interface components

Solution:

- Open Power BI Desktop
- Import the Sales dataset
- Click Home to Transform Data
- Observe the Query Editor window
- identify:
 - Ribbon
 - Query Pane
 - Data Preview
 - Formula Bar
 - Query Setting
 - Applied Steps

1.2 Locate the Query Pane and identify all imported queries

Solution:

- Open Power Query Editor
- Locate the Query Pane on the left side
- Review every imported query
- Select each query individually
- Observe the data preview for each query

1.3 Display the Formula Bar and examine the transformation formula generated by Power Query.

Solution:

- Open Power Query Editor
- Select the View tab
- Enable the Formula Bar
- Select any Applied step
- Observe the generated M expression

1.4 Review the Applied Steps for a query after performing several transformations.

Solution:

- Open Power Query Editor
- Select a query
- Remove one unnecessary column
- Change a data type
- Rename a column
- Review the Applied Steps pane

1.5 Use the Data Preview area to inspect imported data before applying transformation

Solution:

- Open Power Query Editor
- Select the Customer query
- Review the Data preview
- Verify:
 - Column names
 - Data types
 - Blank values
 - Duplicate rows
 - Missing records

1.6 Rename imported queries to meaningful business names

Solution:

- Open Power Query Editor
- Locate the Query Pane
- Right-click the query
- Select Rename
- Rename the queries as:
 - Sales
 - Customers
 - Products
 - Stores
 - Calendar

1.7 Display the Query Setting pane and identify its sections

Solution:

- Open Power Query Editor
- Click the view tab
- Ensure Query Settings is enabled
- Observe:
 - Properties
 - Applied Steps

1.8 Refresh the Data preview after modifying the source dataset

Solution:

- Open Power Query Editor
- Select the desired query
- Click Home then Refresh Preview
- Wait for the preview to reload
- Verify the latest records

1.9 View the complete transformation script using the Advanced Editor

Solution:

- Open Power Query Editor
- Select a query
- Click Home then Advanced Editor
- Review the complete M script
- Close the editor without making changes

1.10 Close Power Query Editor and apply all completed transformations to the Power BI data model

Solution:

- Verify all Applied Steps
- Save the query
- Click Home then Close and Apply
- Wait until Power BI loads the transformed data
- Verify that the tables appear in the Data and Model views.

1.11 View the Query Dependencies diagram to understand how multiple queries are related.

Solution:

- Open Power Query Editor
- Select the view tab.
- Click Query Dependencies.
- Observe the dependency diagram.
- Identify source queries and dependent queries.

1.12 Create a duplicate of an existing query for independent modifications.

Solution:

- Open Power Query Editor.
- Right-Click the Sales Query.
- select duplicate
- Rename the new query as Sales Summary
- Verify that both queries exist independently.

1.13 Create a reference query from an existing Sales query.

Solution:

- Open Power Query Editor.
- Right-click the Sales query.
- Select Reference.
- Rename the new query as Finance Sales.
- Verify that the new query references the original query.

1.14 Display Column Quality indicators to evaluate data quality.

Solution:

- Open Power Query Editor
- Select the view tab.
- Enable Column Quality
- Review each column's quality indicators.

1.15 Display Column Profile statistics for an imported dataset.

Solution:

- Open Power Query Editor
- Select the View tab
- Enable Column Profile
- Select different columns.
- Review statistical information such as:
 - Count
 - Distinct Values
 - Unique Values
 - Empty Values
 - Error Values
 - Minimum
 - Maximum

1.16 Display the Column Distribution visualization to analyze the frequency of values in a dataset.

Solution:

- Open Power Query Editor
- In the Queries pane, select: Customers_Raw
- Click the View tab
- Enable Column Distribution
- Select the city column
- Observe the frequency distribution.

1.17 Rename multiple queries using a consistent naming convention

Solution:

- Go to the Queries pane
- Right-click Sales_Raw
- Select Rename
- Enter FactSales.
- Repeat for Customers_Raw to DimCustomer
- Repeat for Product_Raw to DimProduct

1.18 Create folders to organize queries inside Power Query Editor

Solution:

- Open Power Query Editor
- Right-click inside the query pane
- Select New Group
- Create groups such as:
 - Staging
 - Dimensions
 - Facts
 - Lookup
 - Archive
- Drag related queries into the appropriate groups.

1.19 Identify transformation errors using the Applied Steps pane.

Solution:

- In the queries, select: Error_Data
- On the right side, find Query Settings to Applied Steps
- Click each Applied step one at a time, starting from the first step
- Look at the Data Preview after each step
- Your goal is to identify the step where an error appears.

1.20 Review the entire Power Query project and verify that it is ready for Data Cleaning and Transformation.

Solution:

- Verify all imported queries
- Review query names
- Confirm query groups
- Verify Data Preview
- Review Applied Steps
- Check Query Dependencies
- Enable data profiling tools
- Verify Formula Bar Visibility
- Confirm Query Settings
- Save your Power BI project.

1.21 Remove duplicate customer records from the Customer dataset using Power Query Editor.

Solution:

- In the Queries pane, click Customers_Raw.
- Select the Customer ID columns
- Go to the Home tab
- Click Remove Rows
- Select Remove Duplicates
- Check the result

1.22 Remove blank rows from the Sales dataset before loading it into Power BI.

Solution:

- In the Queries pane, select: Sales_Raw
- Go to the Home tab
- Click Remove Rows
- Select: Remove Blank Rows
- Check the Data Review
- Apply the changes.

1.23 Change the data type of the Order Date column from Text to Date.

Solution:

- Select the Sales_Raw Query
- Find the order Date column
- Select the column
- Go to Transform then Data Type then Date

1.24 Remove unnecessary columns from the Product dataset before loading it into the data model.

Solution:

- Select the Products.Raw Query
- Identify the unnecessary/technical columns
- In this dataset, the main one to remove are: Technical Code, Archive Flag .
- Select the unwanted columns/s
- Go to Home then Remove Columns then Remove Columns

1.25 Filter the Sales dataset to include only records for the year 2025

Solution:

- Select Sales.Raw in the Queries pane
- Locate the Order Date column
- Click the filter row on Order Date
- Choose Date Filters
- Choose Custom Filters
- Set: Order Date is greater than or equal to 01-Jan-2025 AND Order Date is less than or equal to 31-Dec-2025
- Click OK

1.26 Sort the Product dataset alphabetically by Product Name

Solution:

- One Power Query Editor
- Select the Product query
- Select the product name column
- Click the filter/drop-down arrow on Product Name
- Choose Sort Ascending
- Verify that the products are now arranged A to Z

1.27 Replace all null values in the Discount column with zero.

Solution:

- Select the Sales Query
- Select the Discount column
- Go to Transform to Replace values
- In the dialog:
 - Value to Find:null
 - Replace with: 0
- Click Ok
- Check the Discount column to confirm that the blank/null values have become 0.

1.28 Split the Full Name column into First Name and Last Name

Solution:

- In the Queries pane, select the Employee query
- Select the Full Name column
- Go to Transform to Split Column to By Delimiter
- In the delimiter window, choose: space()
- Make sure it splits at the delimiter appropriately
- Click OK

1.29 Merge the City and State columns into a single location column.

Solution:

- In the Queries pane, select the Customer query.
- Select the City column
- Hold Ctrl and select the State column
- Go to: Transform to Merge Columns
- In the dialog, set: Separator to Comma
- Set New column name to:Location
- Click Ok

1.30 Trim extra Spaces from the Customer Name Column.

Solution:

- Select The Customer.Raw
- Select the Customer Name Column
- Go to:
 - Transform then Format then Trim
- Review the values.

1.31 Convert Product Name to UPPERCASE

Solution:

- In the Queries pane, select the Products.Raw.
- Select the Product Name column
- Go to : Transform then Format then UPPERCASE
- Check the transformed values.

1.32 Create an Index Column Starting from 1

Solution:

- Select Customers.Raw
- Go to the Add Column tab
- Click Index Column
- Select: From 1
- Power Query will add a new column called something like Index.

1.33 Create a Conditional Column that classifies sales as High or Low based on Sales Amount.

Solution:

- Select Sales.Raw
- Go to the Add Column tab
- Click Conditional Column
- Set the new column name to : Sales Category
- Create this rule: If Sales Amount is greater than 50000 then High Sales
- Set the Else Value to: Low Sales

1.34 Create a Custom Column that calculates Net Sales Amount minus Discount

Solution:

- Select Sales.Raw
- Go to the Add Column tab
- Click Custom Column
- In New column name, enter: Net Sales
- In the formula box, enter: [Sales Amount]-[Discount]
- Click Ok

1.35 Fill missing values in the Region column using the value from the previous row.

Solution:

- Select Sales.Raw
- Find the Region Column
- Select the Region column
- Go to: Transform then Fill then Down
- Check the result.

1.36 Extract the first four characters from the Product Code column

Solution:

- Select Products.Raw
- Select the product code column
- Go to: Transform then Extract then First Characters
- Enter: 4
- Click OK

1.37 Replace all occurrences of the word "Pending" with "In Progress" in the Order Status column

Solution:

- Select the Orders.Raw
- Select the Order Status column
- Go to: Transform then Replace values
- In the dialog enter:
 - Value to Find: Pending
 - Replace with: In Progress
- Click Ok

1.38 Create a new Data column containing only the Year from the Order Date column

Solution:

- Select the Sales.Raw
- Select the Order Date column
- Go to: Add Column then Date then Year then Year
- Power Query will create a new column containing the year
- Rename the new column: Order Year

1.39 Append two yearly Sales tables (Sales_2024 and Sales_2025) into a single consolidated Sales table.

Solution:

- Open Power Query Editor
- Make Sure both queries are available: Sales 2024 and Sales 2025. They should have the same/similar column structure.
- Select Home then Append Queries
- Select Sales_2024 and Sales_2025
- Verify the total number of rows
- Click Close and Apply

1.40 Merge the Sales table with the customer table to retrieve customer Name for each sales transaction.

Solution:

- Select the Sales query
- Click Home then Merge Queries
- Select the Customer table
- Match the Customer ID column in both tables
- Choose Left Outer Join
- Expand the merged table
- Select only customer Name
- Apply the changes

1.41 Unpivot monthly sales columns (January to December) into a normalized table.

Solution:

- Select Monthly_Sales Query
- Select all 12 monthly columns.
- Go to: Transform then Unpivot Columns then Unpivot Columns
- Power Query will create two columns: Attribute and Values
- Rename: Attribute to Month and Value to Sales Amount

1.42 Pivot the Product Category column to create separate sales columns for each category.

Solution:

- Select Category_Sales Query
- Select the product category column
- Click Transform then Pivot Column
- In the Pivot Column dialog: Values Column is Sales Amount
- Under Advanced options then aggregate value function, select : sum
- Click ok

1.43 Create a reusable function that converts all text values in selected columns to proper case.

Solution:

- Create a Blank Query: Home then New Source then Blank Query and named it: ProperCaseFunction
- Home then Advanced Editor and entered:
(text as text) as text =>
. Text.Proper(text)
 - (text as text)-the function accepts a text value
 - as text - the function returns text
 - Text.Proper(text) - Converts the supplied text to Proper Case.
- You then tested it with: ProperCaseFunction("rAVINDRA pAL")
AND Power Query returned: Ravindra Pal